



UBC
AGROBOT



Bladder Buddy

Catch health issues early for a better tomorrow.

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Overview

01

Problem

02

Solution

03

pH Sensor

03

Protein Sensor

03

Benefits

Dehydration



Infections

“Healthy people, brighter future.”

💡 1 in 7 American adults have kidney disease, and up to 90% are unaware.

Kidney Functions

Metabolism

PROBLEM

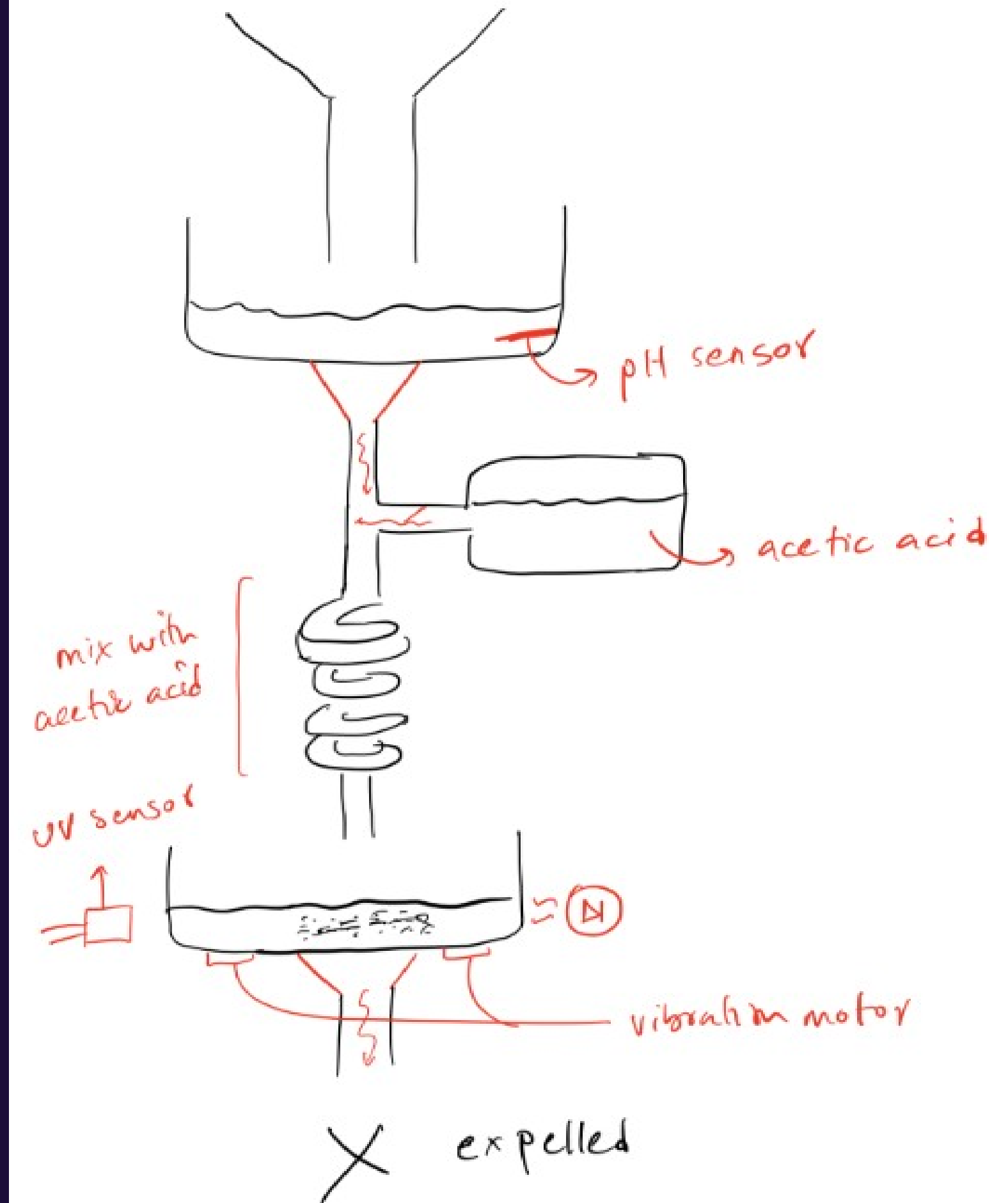
Currently, it's inefficient to monitor personal health:

- People **miss early signs** of health issues
 - **testing is inconvenient**
 - **tests not done regularly**
- Getting results takes **time**
- Doctors'/Labs' time could be spent on other issues

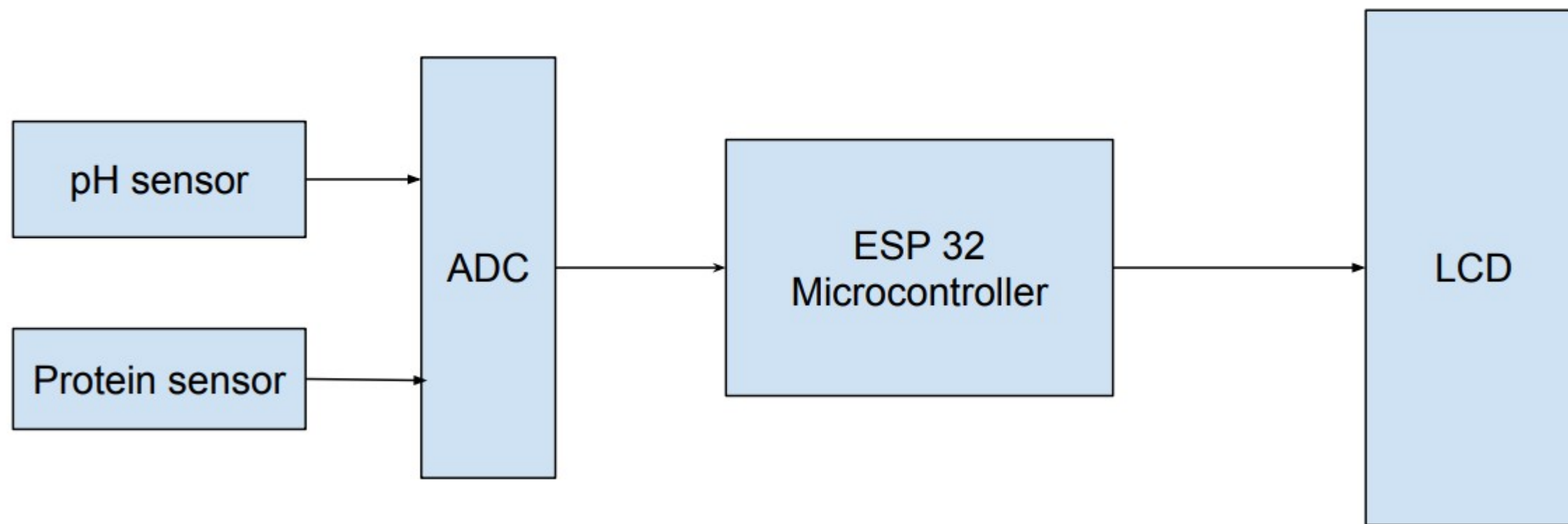
Solution

BLADDER BUDDY

- Urine detection system
 - pH sensor
 - protein sensor



Block Diagrams



pH Sensor

pH is good warning sign for possible problems.

Glass probe measures voltage/charge caused by hydrogen ion activity.



pH 4502C Sensor & BNC Probe
+/- 0.1 pH

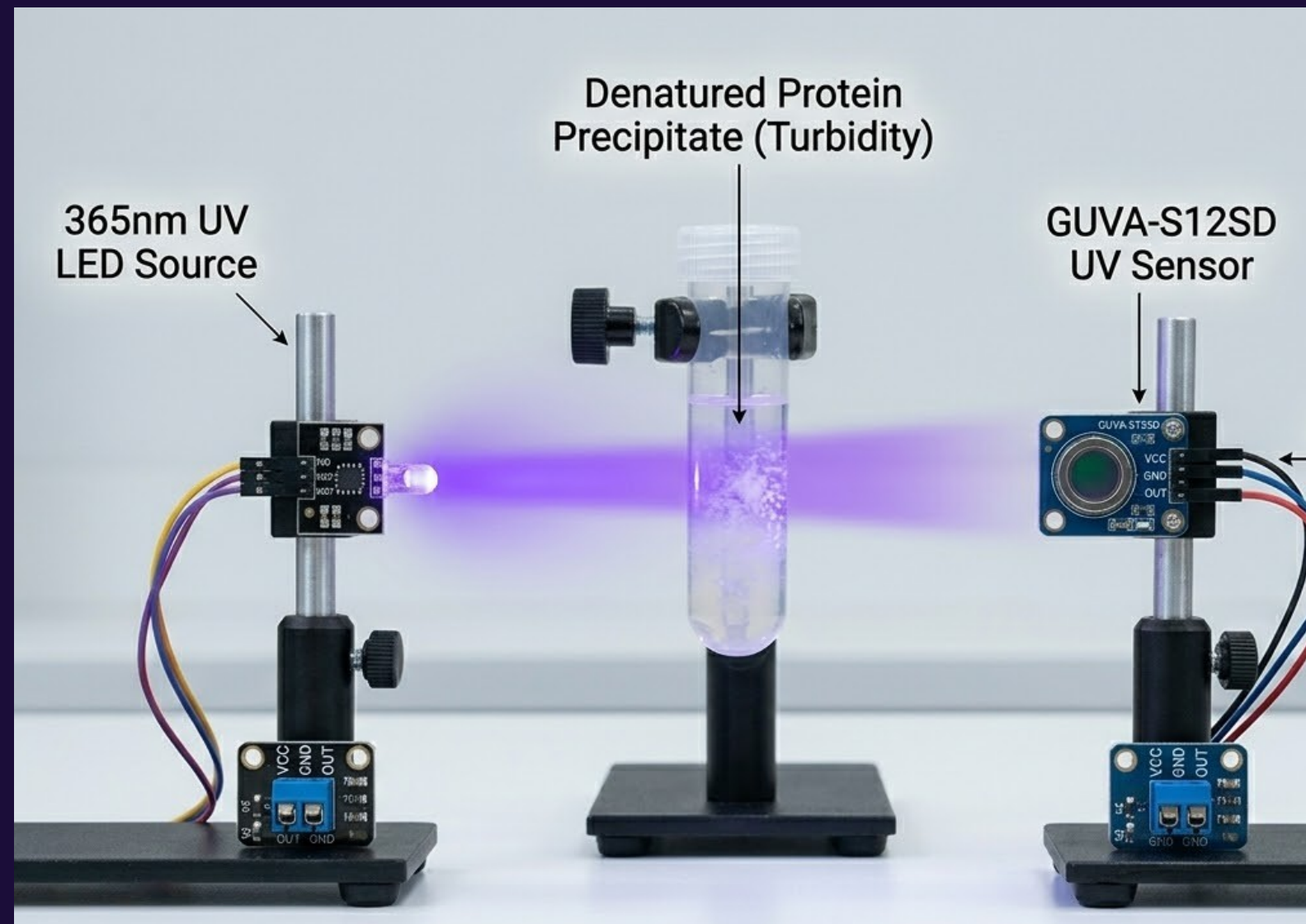
pH Indications:

5.5 - 7.0 - Healthy

< 5.5 (acidic) - dehydrated
risk of kidney stones
Metabolic issues (acidosis)

> 7.0 (basic) - Urinary tract infection (UTI)
Kidney issues

Protein Sensor



Acetic Acid denatures protein

Urine becomes cloudy

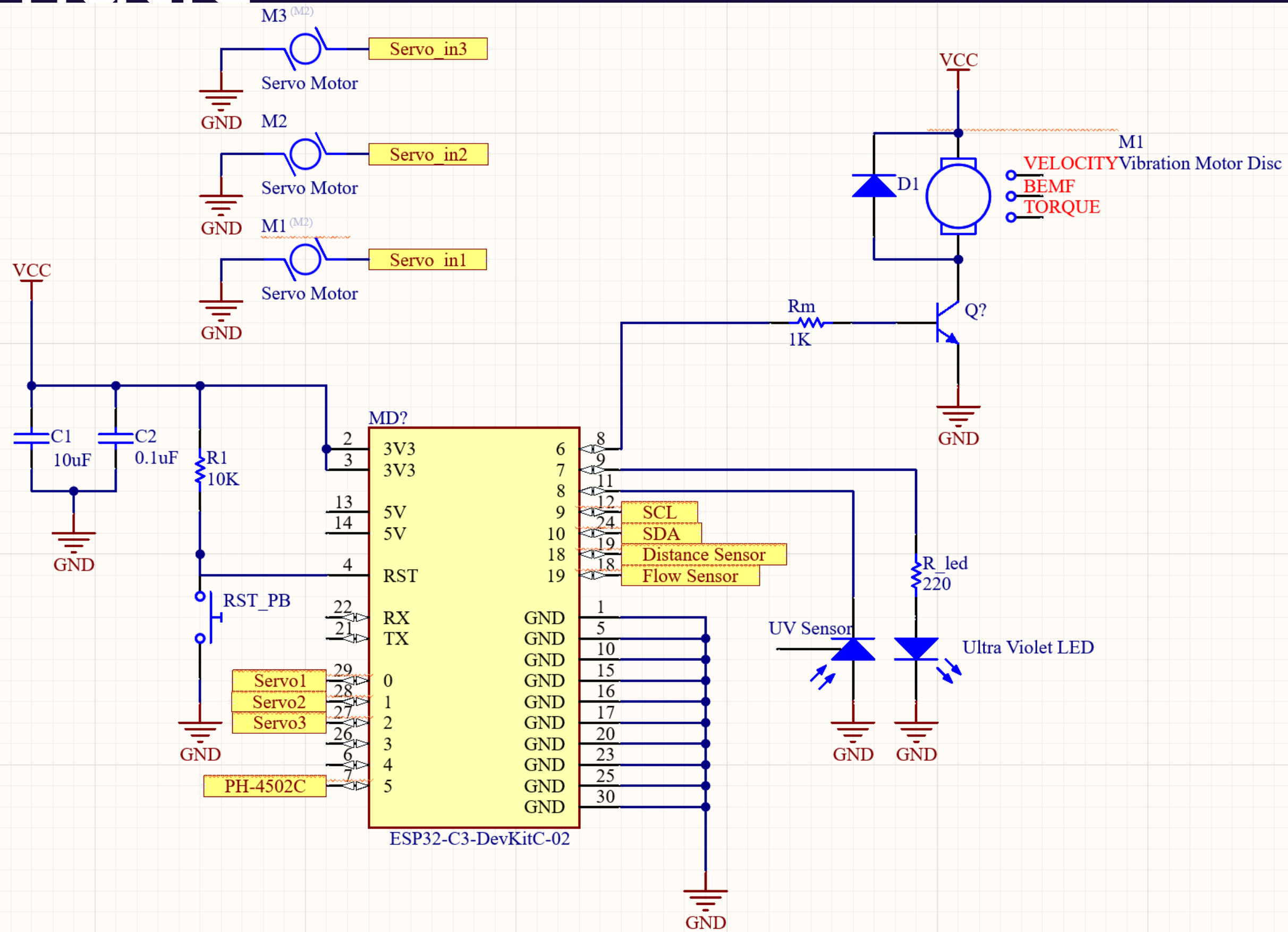
365 nm light is primarily scattered by protein in urine sample

UV sensor records the amount of light passing through and converts it to a voltage

Protein Sensor

- **High voltage (3.3 V): No protein**
- **Medium voltage (1.5 - 2.5 V): Trace proteinuria**
- **Low voltage (<1.0 V): Seek doctor consultation**

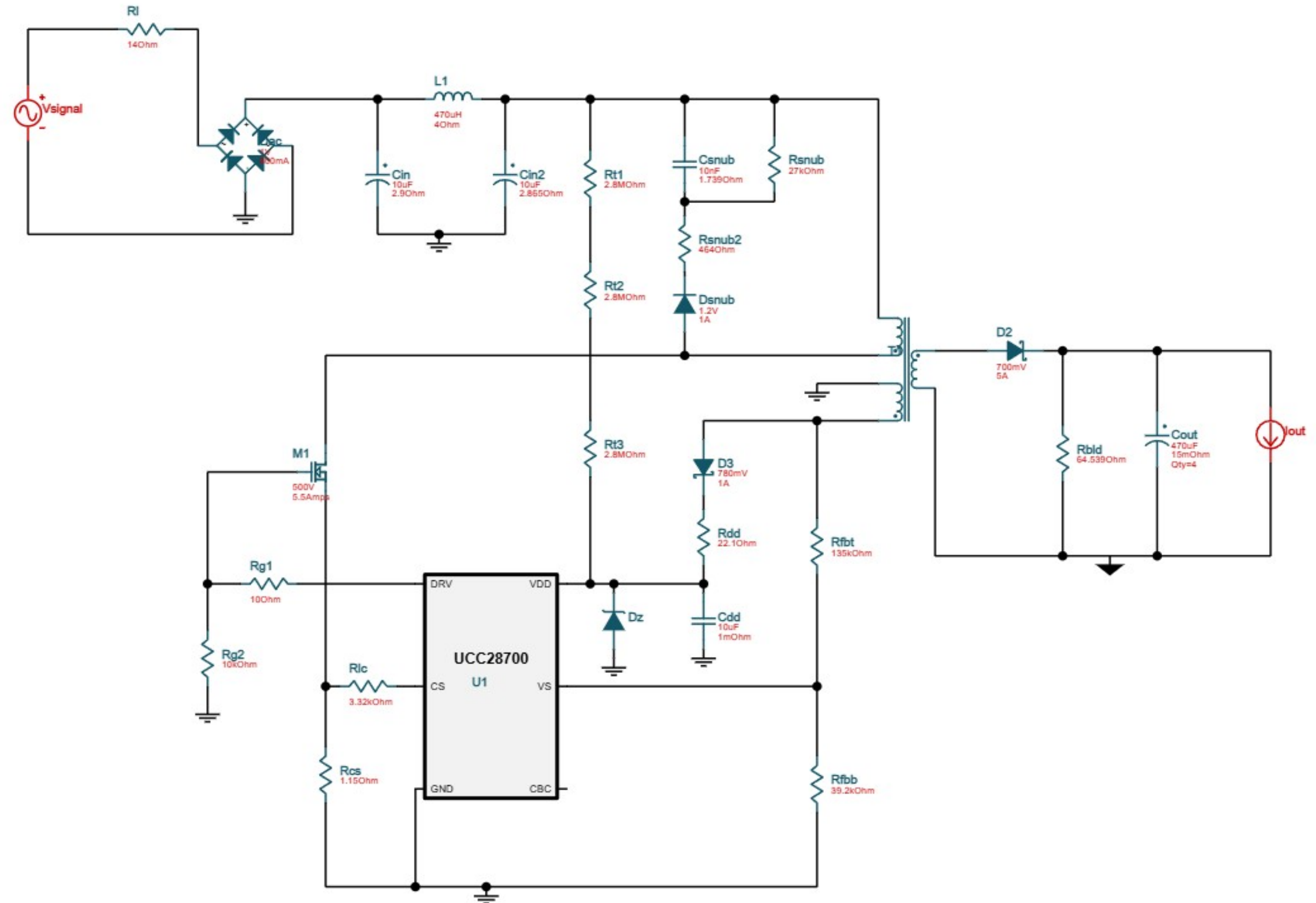
Schematic



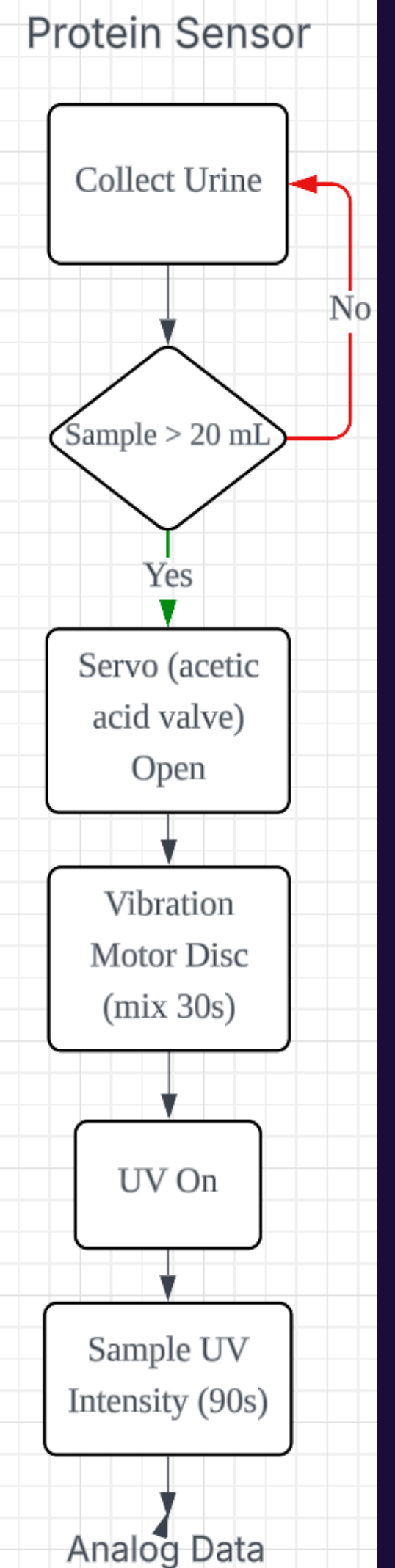
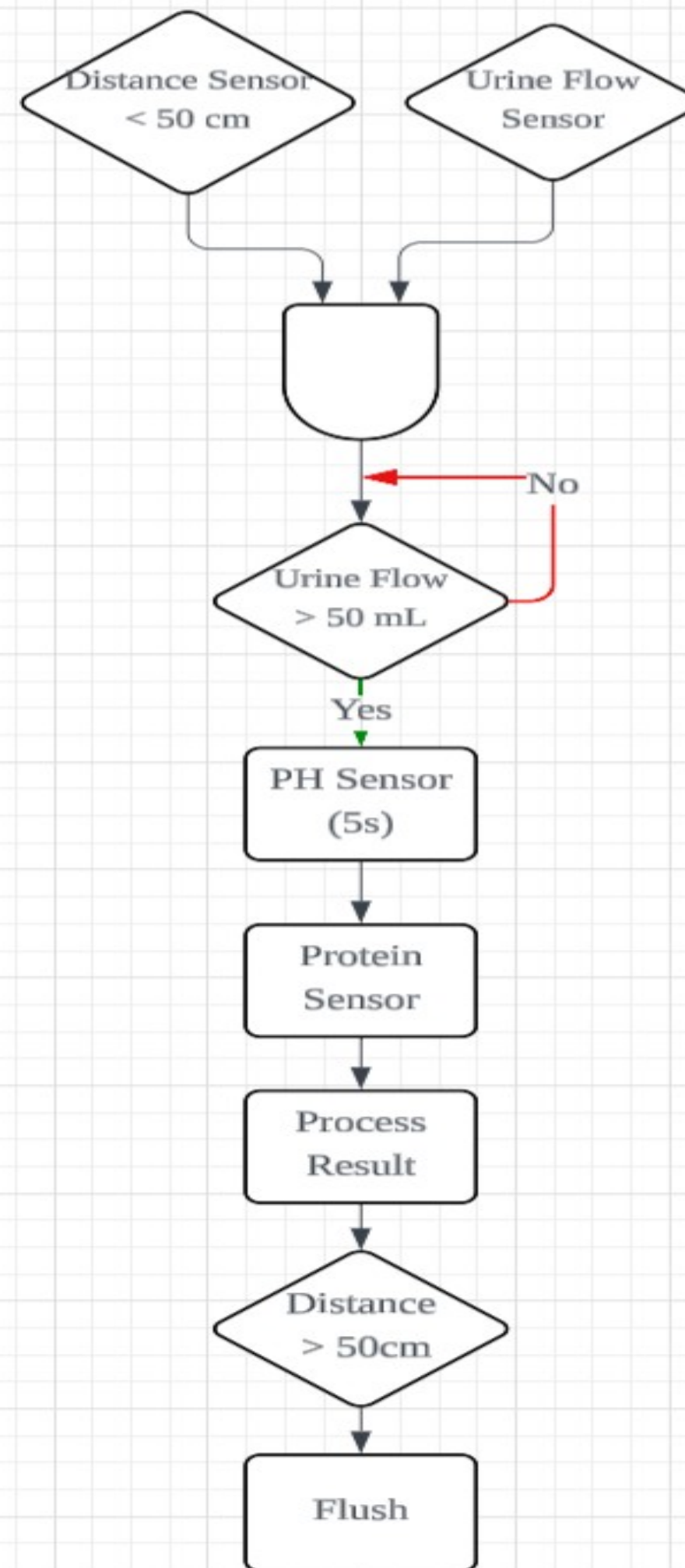
Voltage Converter Schematic

Input: 120V AC
Output: 5V, 1A

Next Step:
EMI Circuit Design
Spice Simulation
PCB Design

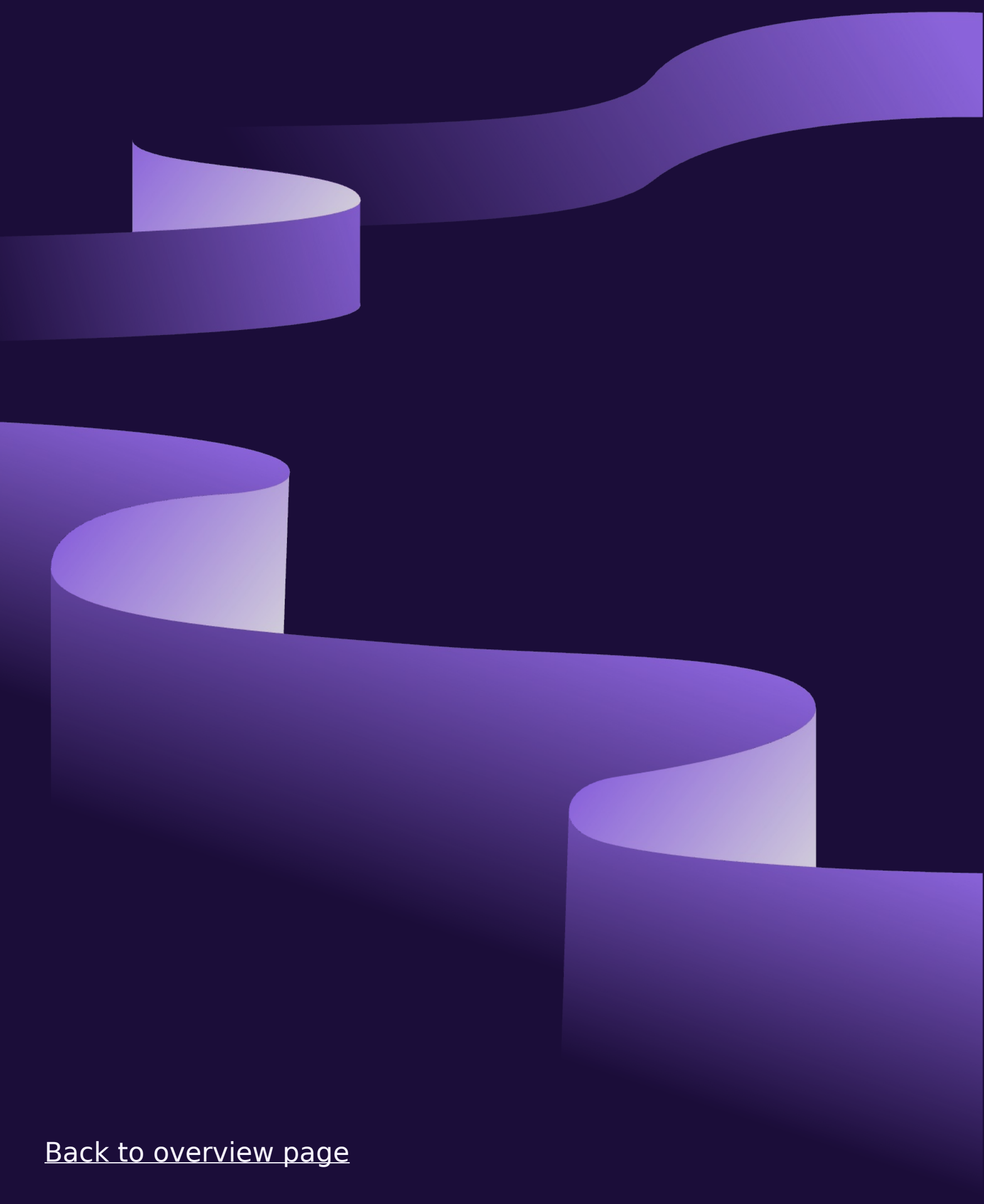


Software Block Diagram



Component Pricing

	Price (\$)	Total
ESP 32	9.38	122.61
pH sensor	27.53	
Protein sensor	35	
Mg90 Servo Motor	30	
Flow Sensor	12	
Distance Sensor	8.7	



Benefits

- **Early disease detection**
- **Reduced need for laboratory analysis of urine**
- **No need for plastic waste**
- **Long-term economic and medical efficiency**

Conclusion

- **Cost effective**
- **Reduces disposable plastic**
- **Quick diagnosis for long-term health problems**

Thank You!